

1 a) ± 1 . All q have length 1. ✓

b) ± 2 ✓

c) ± 1 . Both are ± 1 ✓

2 a)
$$\begin{bmatrix} +1 & 0 \\ \vdots & \vdots \\ +1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -10 \\ \vdots & -9 \\ \vdots & \vdots \\ 1 & 10 \end{bmatrix} \hat{x} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \quad (1^2 + \dots + 10^2 = \alpha)$$

$$\begin{bmatrix} 2 & 0 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \end{bmatrix}$$

$$C = \frac{1}{2}$$

$$D = 0$$

b)
$$C_1 \begin{bmatrix} 1 \\ \vdots \\ \vdots \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} -10 \\ -9 \\ \vdots \\ \vdots \\ +9 \\ +10 \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ \vdots \\ 2 \\ \vdots \\ \vdots \end{bmatrix}$$
 ✓

Subject:

Year:

Month:

Date:

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3 a) $Q (Q^T Q)^{-1} Q^T = P_Q$ ✓

$$A(A^T A)^{-1} A^T = P_A$$

b) yes. their column space is equal. ✓

$$P_Q Q = Q \quad \det(P_Q) = 0$$

c) 3 ✓

4 a) if x repeated in other 9, x^4

else x^{42} ✓

b) $x, 0, \frac{1}{2}$ ✓